# ECON 0150 | Economic Data Analysis

*The economist's data analysis skillset.*

## *Part 2 | Wrap Up*

# The Part 2 Toolkit

*Three figures, each matched to a set of variables*

| Part | Variables | Figure | Later becomes… |
|---|---|---|---|
| 2.1 | Numerical by Numerical | Scatterplot | Regression (Part 4) |
| 2.2 | Numerical by Categorical | Boxplot (+ stripplot) | Two-sample test (Part 3) |
| 2.3 | Numerical by Numerical by Categorical | Scatter colored by category | Regression with a control (Part 5) |

Before we draw, we prepare: **filter** rows that don't belong, **log** skewed variables, **merge** data from different sources.

# Choose the Visualization

# Exercise 1: Commuting and Gas

*What are the variable types, and what figure would you draw?*

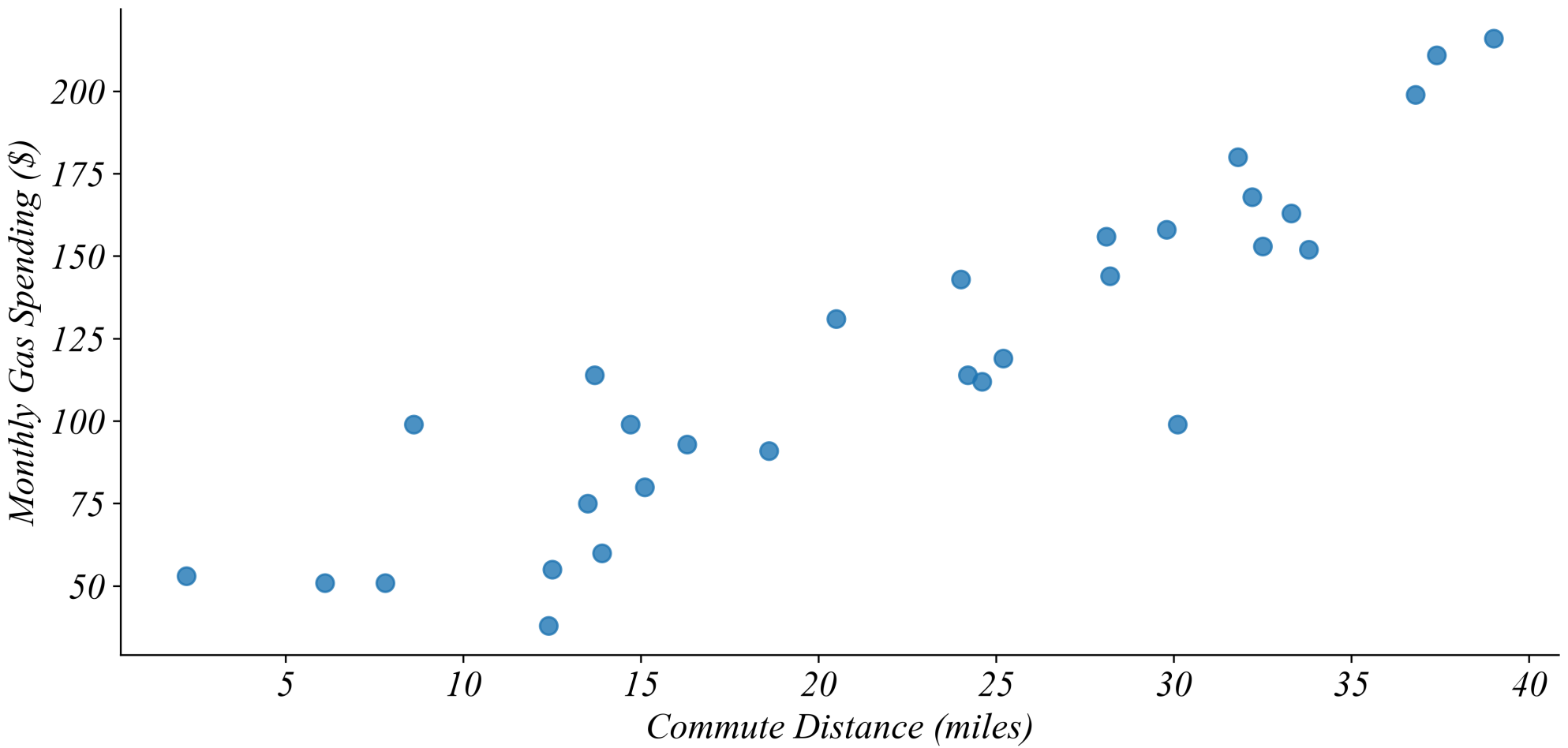
A dataset on commute distance and monthly gas spending for 30 workers.

| Commute_Miles | Gas_Spend |
|---|---|
| 2.2 | 53 |
| 37.4 | 211 |
| 31.8 | 180 |
| 14.7 | 99 |
| 28.1 | 156 |

**Variable Types:** Numerical by Numerical

# Exercise 1: Commuting and Gas

*What are the variable types, and what figure would you draw?*




A ***Scatterplot*** shows how one numerical variable moves with another.

# Exercise 2: Rent by Area

*What are the variable types, and what figure would you draw?*

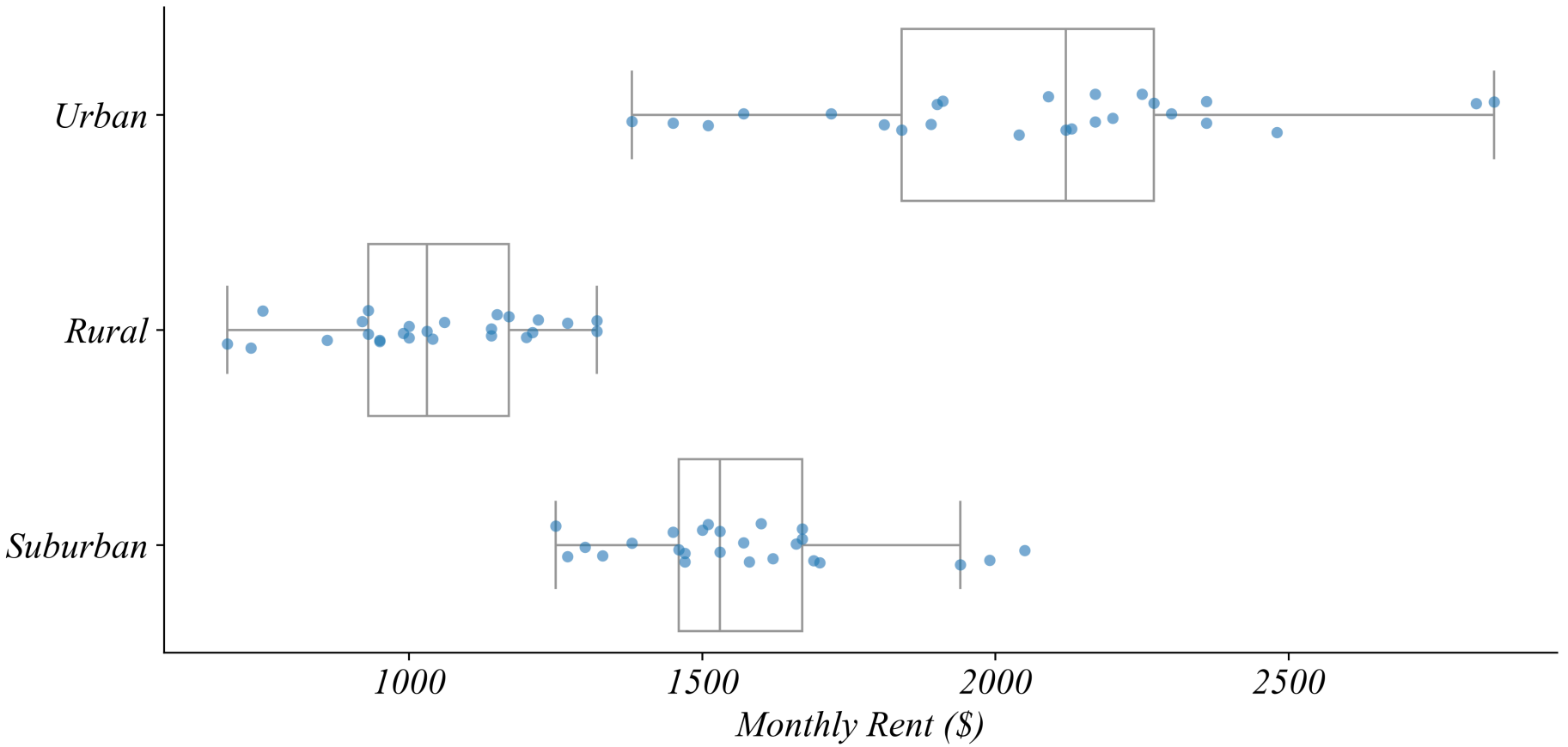
A dataset on monthly rent for 75 apartments.

| Area | Rent |
|---|---|
| Urban | 2130 |
| Rural | 1060 |
| Urban | 1720 |
| Rural | 860 |
| Rural | 950 |

**Variable Types:** Numerical by Categorical

# Exercise 2: Rent by Area

*What are the variable types, and what figure would you draw?*




A ***Boxplot by category*** compares the distribution across groups. The ***stripplot*** shows each apartment.

# Exercise 3: Studying and Exam Scores

*What are the variable types, and what figure would you draw?*

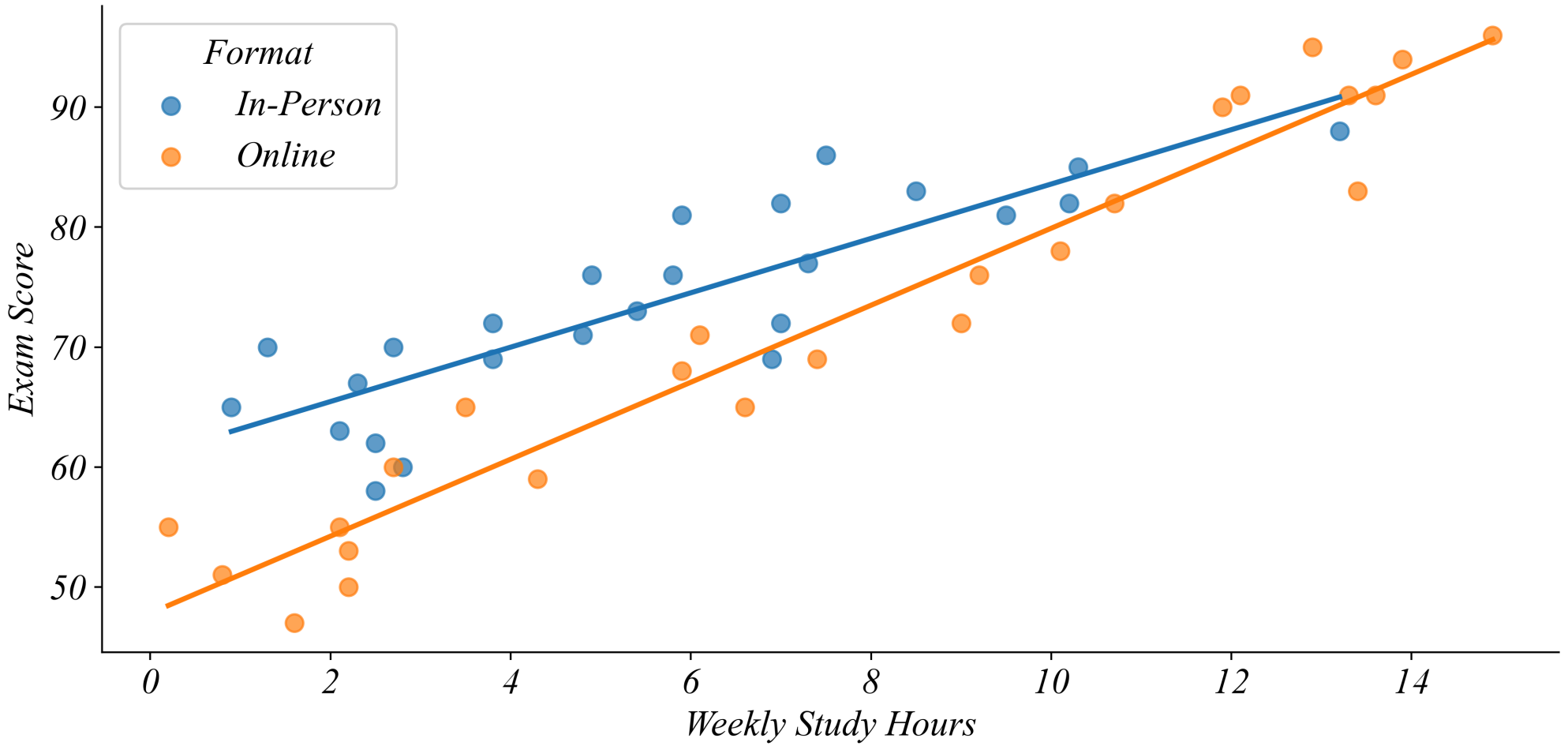
A dataset on weekly study hours and exam scores for students in two course formats.

| Format | Study_Hours | Exam_Score |
|---|---|---|
| Online | 3.5 | 65 |
| Online | 13.4 | 83 |
| Online | 0.8 | 51 |
| In-Person | 0.9 | 65 |
| In-Person | 1.3 | 70 |

**Variable Types:** Numerical by Numerical by Categorical

# Exercise 3: Studying and Exam Scores

*What are the variable types, and what figure would you draw?*




A ***Scatter colored by category*** shows whether the *relationship* differs between groups.

# Exercise 4: Inflation and Unemployment

*What are the variable types, and what figure would you draw?*

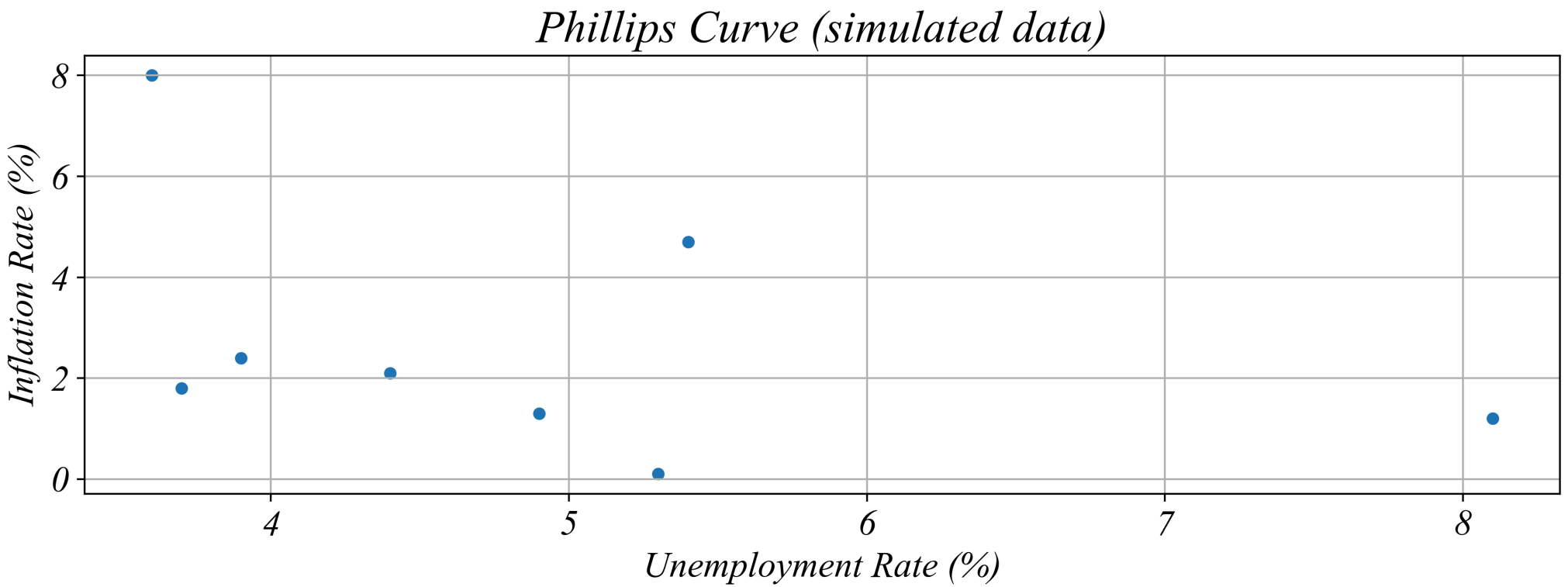
US unemployment and inflation rates each year since 2015.

| year | unemployment | inflation |
|---|---|---|
| 2015 | 5.3 | 0.1 |
| 2016 | 4.9 | 1.3 |
| 2017 | 4.4 | 2.1 |
| 2018 | 3.9 | 2.4 |
| 2019 | 3.7 | 1.8 |

**Variable Types:** Numerical by Numerical, measured over time

# Exercise 4: Inflation and Unemployment

*What are the variable types, and what figure would you draw?*

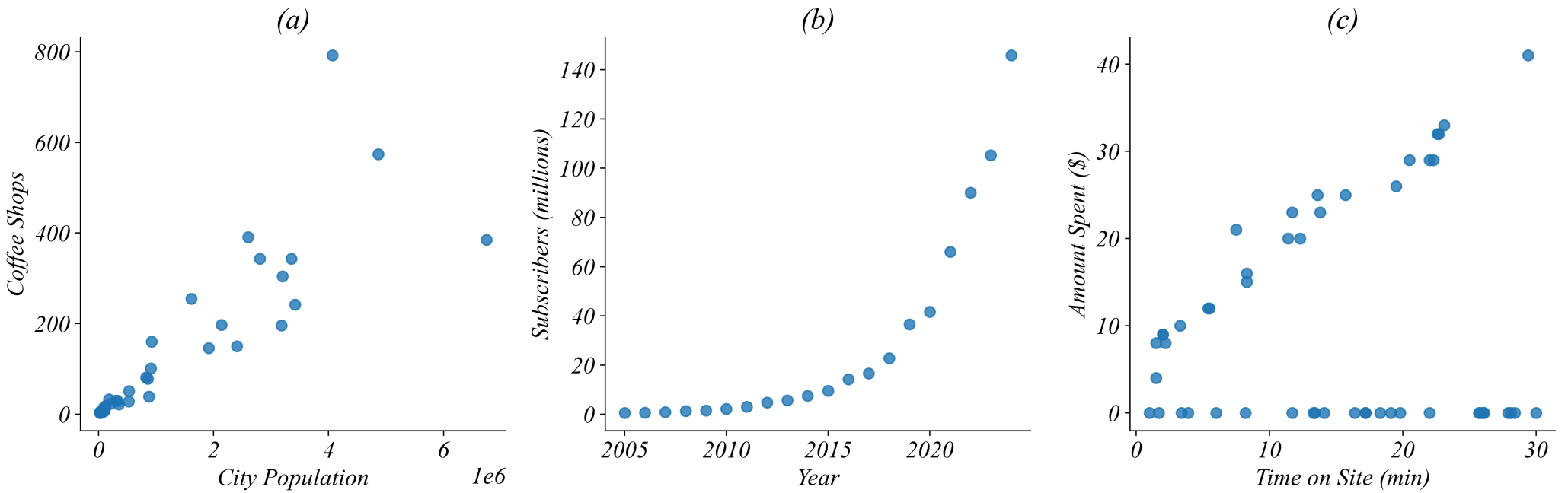



A ***Scatterplot*** works for timeseries too, like coffee and oil prices in Part 2.1. Labeling the points with years keeps the time order.

# Ask a Specific Question

# Ask a Specific Question

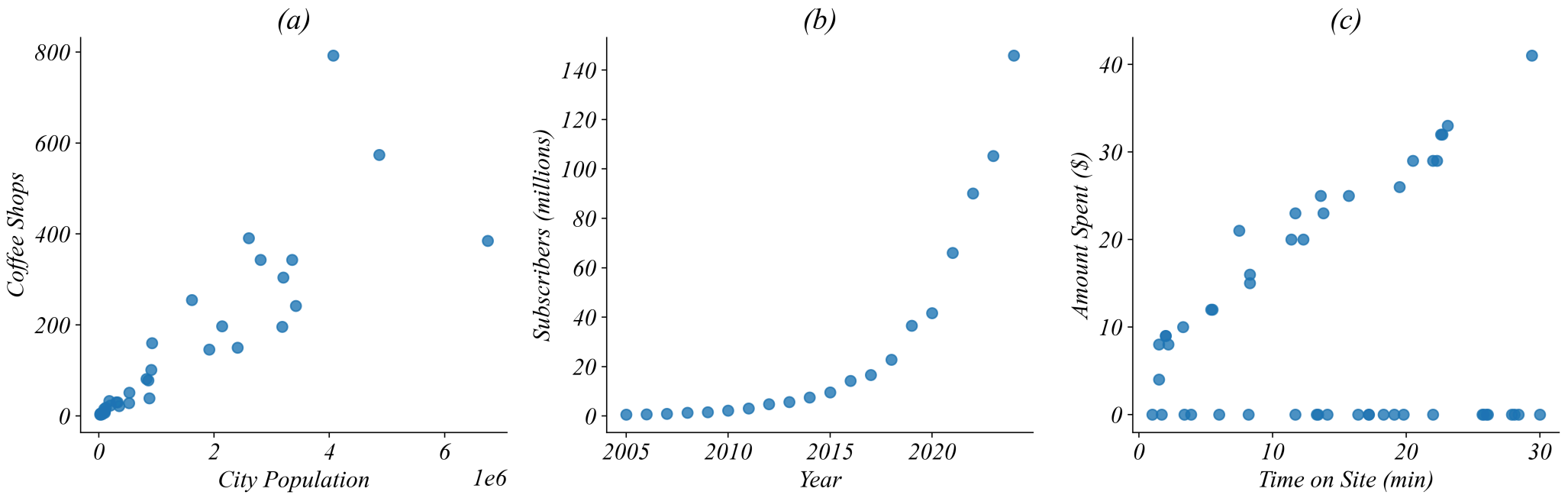
*Each figure shows the data accurately. Does it answer the question?*




- ***(a)*** *How does the number of coffee shops change with city size, for small cities as well as large ones?*
- ***(b)*** *Do subscribers grow at a steady rate, or does the growth rate change over time?*
- ***(c)*** *Among customers who buy something, do longer visits lead to bigger purchases?*

# Ask a Specific Question

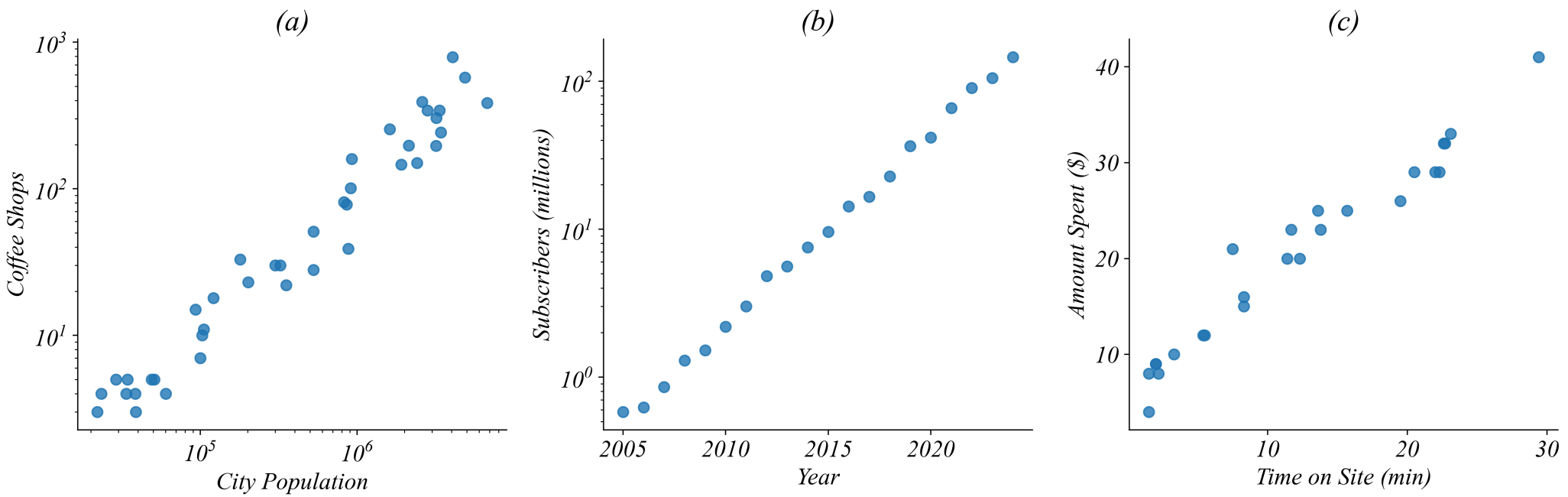
*What does each figure show well, and what does it make hard to see?*




- ***(a)*** *Most cities clump in the bottom-left corner. We can't see the small cities at all.*
- ***(b)*** *The curve is accurate: growth is speeding up. But it's hard to judge by eye whether a curve bends at a steady rate.*
- ***(c)*** *The zeros are real and useful: most visitors buy nothing. But our question is only about the buyers.*

# Ask a Specific Question

*Log both axes, log one axis, or filter*




- ***(a)*** *Log* ***both*** *axes: small and large cities spread out, and each step is a multiple of the last.*
- ***(b)*** *Log the* ***y-axis only****: a steady percentage growth rate becomes a straight line.*
- ***(c) Filter*** *to buyers: among those who spend, longer visits go with bigger purchases.*

# Log Scales

# Reading Log Scales

*Equal distances on a log scale mean equal multiples*

*a. On a log10 scale, each one-unit increase means the original value has been…*

*multiplied by 10*

*b. On a log2 scale, moving up 3 units means the original value has been multiplied by…*

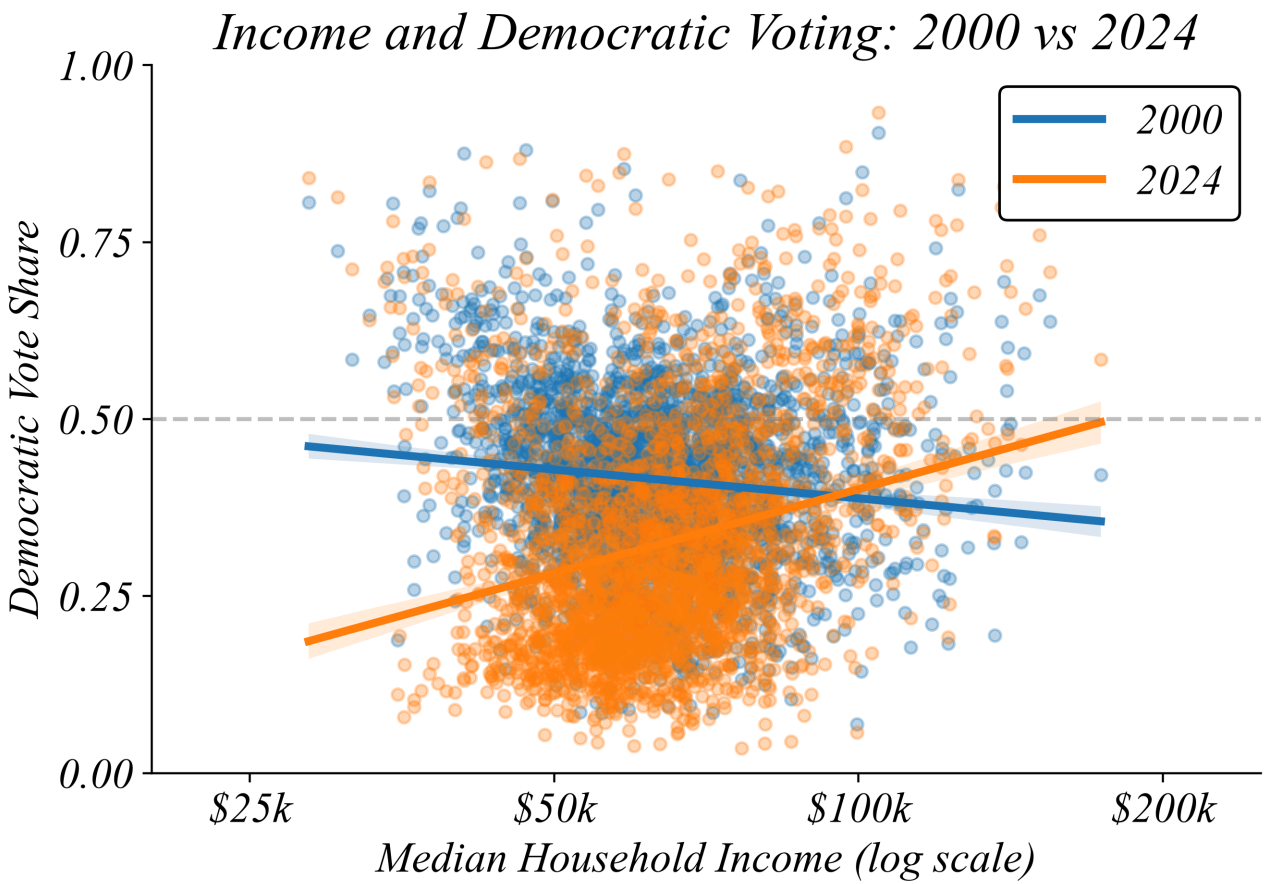
*2 × 2 × 2 = 8*

*c. City A has log10(Population) = 4. City B has log10(Population) = 6. How many times larger is City B?*

*Two units apart on log10: 10 × 10 = 100 times larger*

# Reading Log Scales

*On the income axis in Part 2.3, what does equal spacing mean?*




*$25k → $50k → $100k → $200k: each step to the right is a **doubling** of income.*

# Reading Log Scales

*Why not just log every skewed variable?*

In Part 2.2, most Starbucks rows had zero revenue.

- *log(0) is undefined, so logging a column of zeros fails.*
- *The zeros were a different kind of row: offers viewed, not purchases.*
- ***Filter first, then transform.***

# Investigate Before You Visualize

# Investigate: App Sessions

*A coffee app records every session, whether or not the customer buys.*

| Session | Event | Minutes | Amount |
|---|---|---|---|
| S01 | Purchased | 4 | 6 |
| S02 | Browsed | 2 | 0 |
| S03 | Purchased | 9 | 11 |
| S04 | Browsed | 7 | 0 |
| S05 | Purchased | 15 | 17 |
| S06 | Browsed | 12 | 0 |
| S07 | Purchased | 6 | 8 |
| S08 | Browsed | 3 | 0 |
| S09 | Purchased | 12 | 14 |
| S10 | Browsed | 10 | 0 |
| S11 | Purchased | 18 | 21 |
| S12 | Browsed | 5 | 0 |

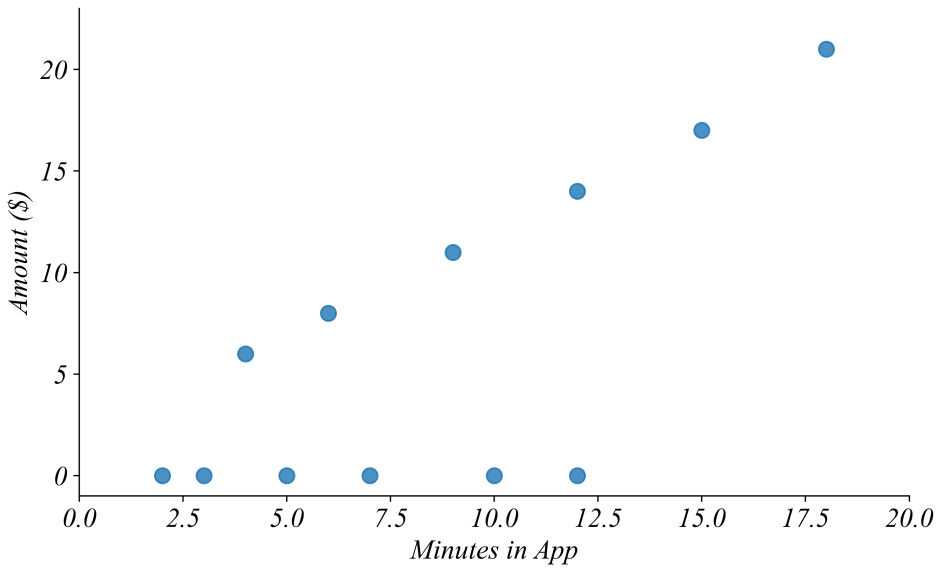
a. *Sketch a scatter of Minutes vs Amount using **all** the data.*

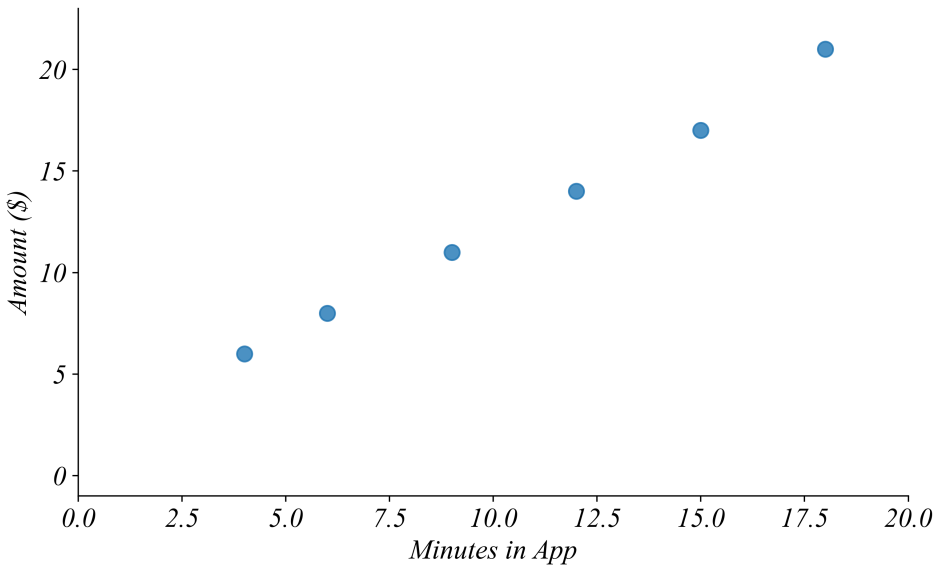
b. *Which rows should we drop to ask "do longer sessions lead to bigger orders?"*

c. *Sketch the scatter again after filtering.*

# Investigate: App Sessions

*All sessions vs. purchases only*




*All rows: half the points sit at zero and hide the pattern.*

*Filter* `Event == 'Purchased'`*: longer sessions, bigger orders.*

# Filter, Group, Summarize

# Practice 1: Trace the Filter

*Which apartments remain after filtering?*

| Apt | Area | Rent | Bedrooms |
|---|---|---|---|
| A1 | Urban | 1850 | 1 |
| A2 | Rural | 900 | 3 |
| A3 | Suburban | 1400 | 2 |
| A4 | Urban | 1450 | 1 |
| A5 | Suburban | 1650 | 3 |
| A6 | Rural | 1200 | 2 |

**Filter:** `((Area == 'Urban') OR (Area == 'Suburban')) AND (Rent < 1500)`

*Answer: A3, A4*

# Practice 2: Build the Filter

*Construct the boolean logic from the pieces*

**Goal:** Find all apartments that are Rural, OR that have at least 2 bedrooms AND rent under $1,500.

*1. (Area == 'Rural')*

*2. (Bedrooms >= 2)*

*3. (Rent < 1500)*

*4. (Area == 'Urban')*

*Answer: 1 OR (2 AND 3)*

*Using the table from Practice 1: A2, A3, A6*

# Practice 3: Filter → Group → Summarize

*The Part 2.2 pipeline, by hand*

| Txn | Offer | Event | Amount |
|---|---|---|---|
| T01 | BOGO 5 | Purchase | 12 |
| T02 | BOGO 10 | Viewed | 0 |
| T03 | BOGO 5 | Purchase | 20 |
| T04 | Discount | Purchase | 8 |
| T05 | BOGO 10 | Purchase | 30 |
| T06 | BOGO 5 | Viewed | 0 |
| T07 | Discount | Purchase | 16 |
| T08 | BOGO 10 | Purchase | 18 |

**Operations:** 1) Filter for `Event == 'Purchase'` 2) Group by Offer 3) Calculate the mean and count of Amount

# Practice 3: Filter → Group → Summarize

*The Part 2.2 pipeline, by hand*

| Offer | Mean Amount | Count |
|---|---|---|
| BOGO 5 | 16 | 2 |
| BOGO 10 | 24 | 2 |
| Discount | 12 | 2 |

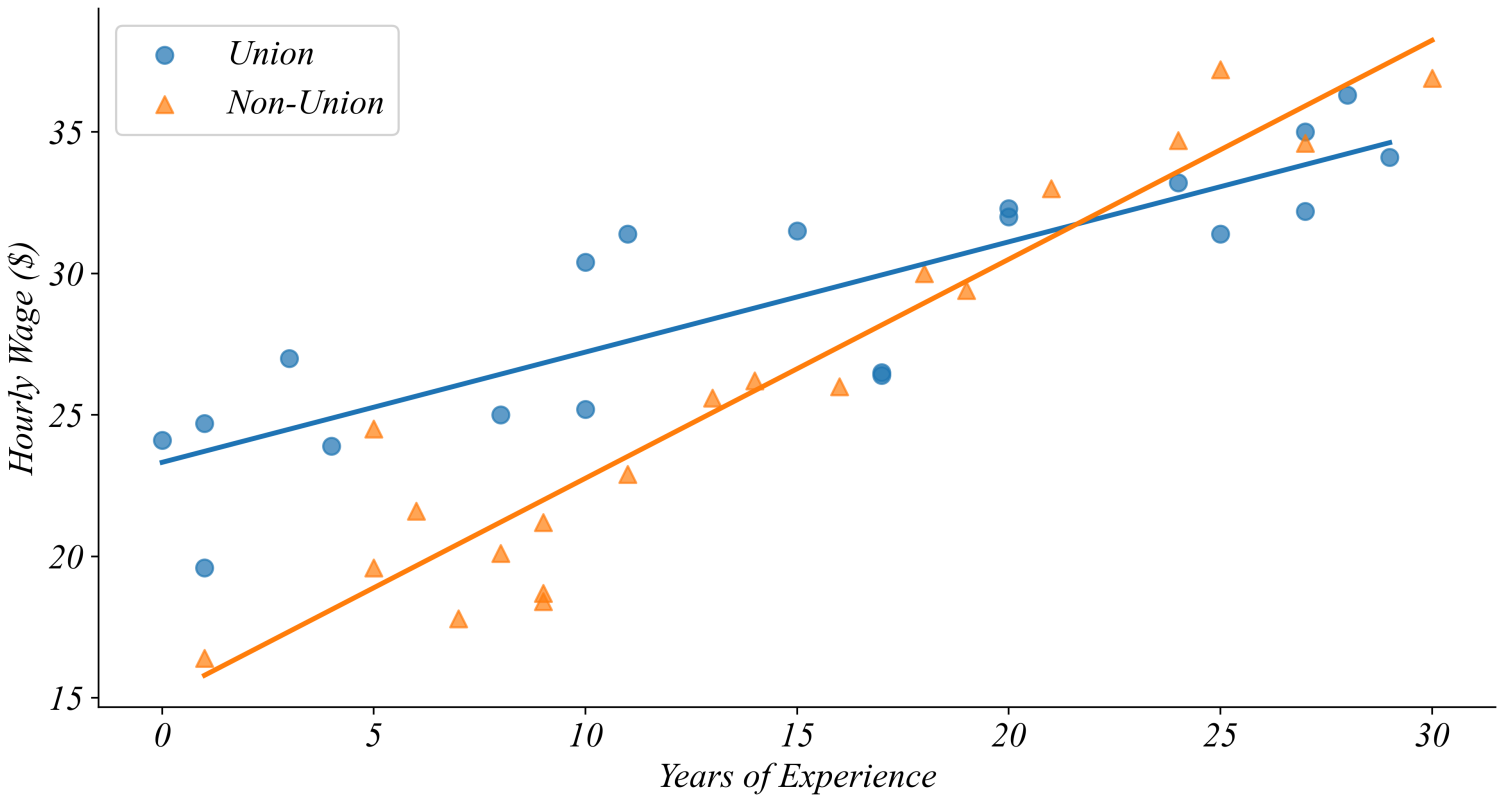
**What if we forgot the filter?** BOGO 5 becomes (12 + 20 + 0) / 3 = 10.67, and BOGO 10 becomes (0 + 30 + 18) / 3 = 16.

*The viewed rows aren't purchases. Leaving them in drags every mean toward zero.*

# Relationships by Category

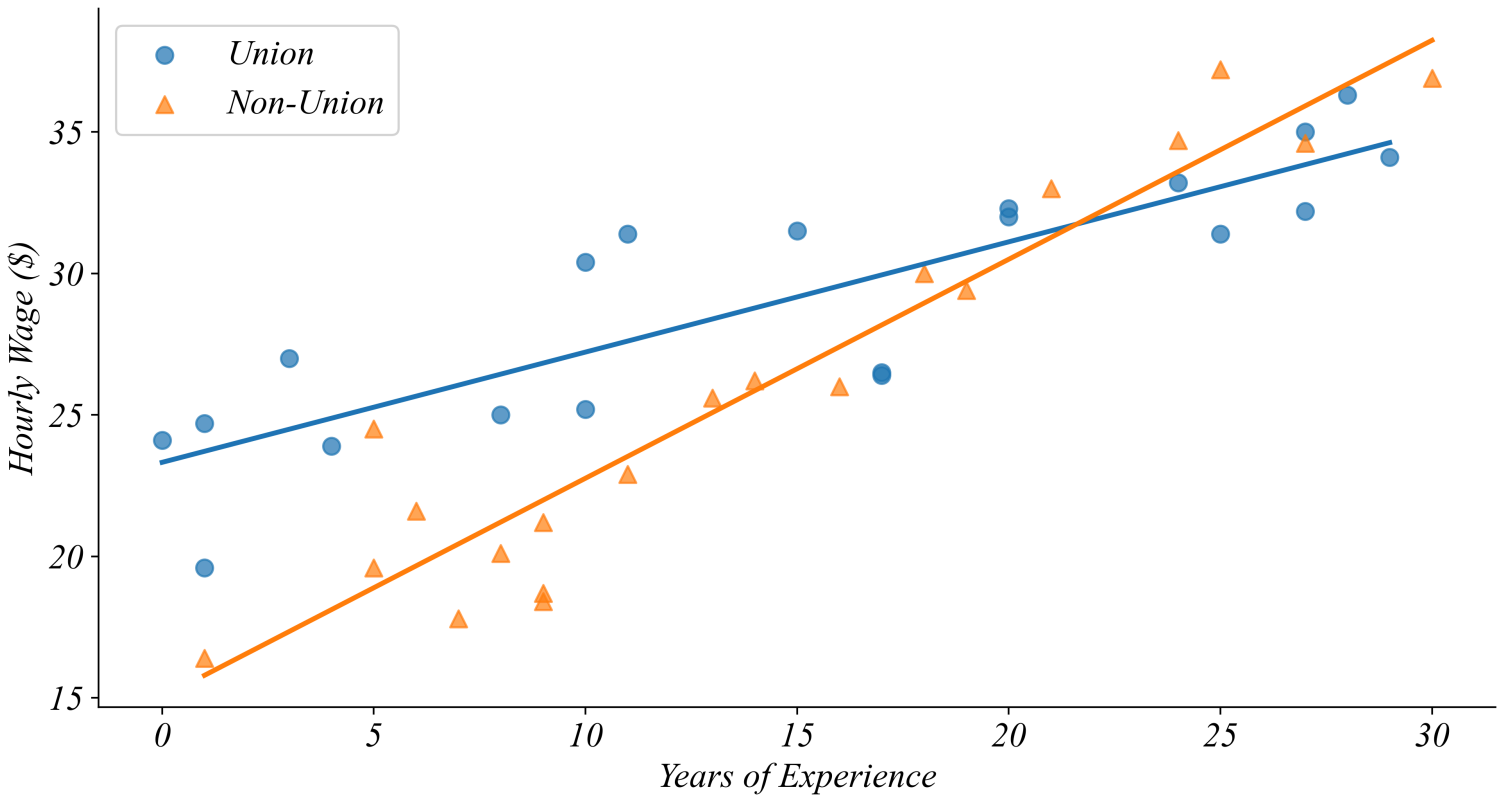
# Read a Scatter by Category

*Hourly wage and experience for union and non-union workers*



# Read a Scatter by Category

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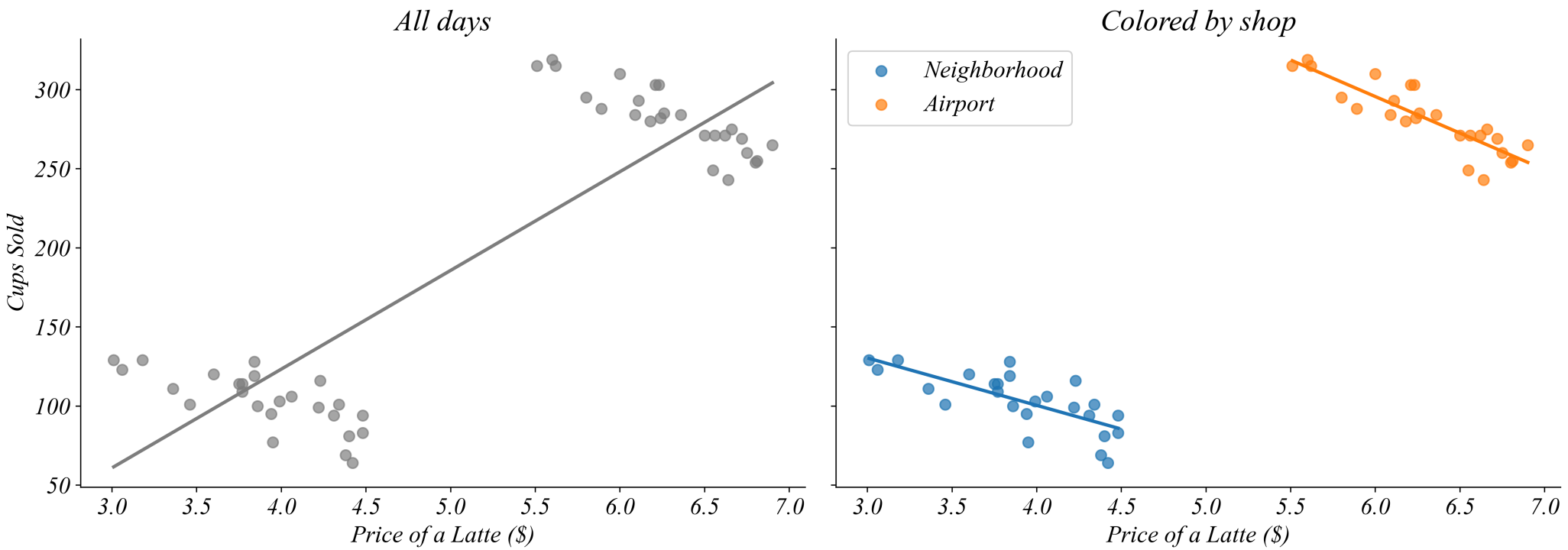



*a. Both increase. b) Union at 5 years; non-union at 28 years.*

*b. Union workers start higher, but non-union wages rise faster with experience (a steeper line).*

# Why Color by Category?

*Each point is one day at one coffee shop*




*All days together: higher prices, more cups sold. Is demand upward sloping?*

*Within each shop, higher prices sell **fewer** cups. The airport just has higher prices **and** more customers.*

# Why Color by Category?

*The same lesson from Part 2.3*

- *Counties shifted Republican on average (boxplot by year)*
- *Yet large, high-income counties vote Democratic (scatter)*
- *And the income relationship flipped between 2000 and 2024 (scatter by year)*
- ***A pattern in pooled data can differ from the pattern within each group.***

# Combining and Reshaping Data

# Practice 4: Trace the Merge

```
pd.merge(population, rents, on='City')
```

**population**

| City | Population |
|---|---|
| Alder | 120,000 |
| Birch | 850,000 |
| Cedar | 2,400,000 |
| Dogwood | 45,000 |

**rents**

| City | Rent_2019 | Rent_2024 |
|---|---|---|
| Alder | 950 | 1150 |
| Birch | 1300 | 1600 |
| Cedar | 2000 | 2300 |
| Elm | 800 | 980 |

How many rows in the merged data? Which cities drop out?

*3 rows. Dogwood and Elm drop out: each appears in only one table.*

# Practice 5: Trace the Melt

*Wide to long, so year can become a category*

**Wide**

| City | Rent_2019 | Rent_2024 |
|---|---|---|
| Alder | 950 | 1150 |
| Birch | 1300 | 1600 |
| Cedar | 2000 | 2300 |

**Long**

| City | Year | Rent |
|---|---|---|
| Alder | 2019 | 950 |
| Birch | 2019 | 1300 |
| Cedar | 2019 | 2000 |
| Alder | 2024 | 1150 |
| Birch | 2024 | 1600 |
| Cedar | 2024 | 2300 |

*3 rows become 6. Now `Year` is a column we can put on an axis or use for color.*

# Putting It Together

# A Research Question

*Do bigger cities have higher rents, and has that changed since 2019?*

Using the Part 2.3 template, which figure answers each question?

1. ***Did rents rise?*** *Boxplot of Rent by Year*
2. ***Do bigger cities charge more?*** *Scatter of log Population vs Rent*
3. ***Has that relationship changed?*** *Scatter of log Population vs Rent, colored by Year*

*Data steps: merge population and rents on City, melt the rent years, log population.*

# Part 2 | Summary

- ***Choose*** *the figure from the variable types: scatter, boxplot by category, or scatter by category*
- ***Filter*** *out rows that don't belong to the question before summarizing or plotting*
- ***Log*** *skewed variables: equal distances become equal multiples (but never log zeros)*
- ***Group*** *to summarize by category: one row per group*
- ***Merge*** *when the question needs variables from two sources;* ***melt*** *to turn columns into a category*